



Thomas Barbosa

DATA SPECIALIST · AI & AUTOMATION

SUMMARY

I lead the Data & AI team at Grupo Raposo Plásticos: 130+ automations and 10 AI agents in production, from the ERP to the decision dashboard. Before that, pricing and market intelligence at Solar Coca-Cola and BI at Arco Educação.

EXPERIENCE

Data Specialist · Grupo Raposo Plásticos

2025 — present

I lead the Data & AI team that brings solutions to the group's plants: an AI agents hub and the group's automation web — ERP and MES reads, agents that answer and act on WhatsApp, image and PDF reports and more than a hundred n8n workflows.

Senior Pricing Analyst · Solar Coca-Cola

2024 — 2025

Market Intelligence Analyst (Mar 2024 — Apr 2025)

Market intelligence and Revenue Growth Management: SQL over Scanntech and Nielsen retail data, ETL from many sources to anticipate revenue risks and opportunities, and Power BI dashboards that guided portfolio pricing.

Projects and BI Analyst · SAS Education Platform · Arco Educação

2020 — 2024

Projects Analyst I (Apr 2021 — Dec 2023) · Assessment Management Intern (Mar 2020 — Apr 2021)

Production data for teaching materials in SQL Server (procedures and SSIS) and BigQuery; productivity, cost and quality dashboards in Power BI and Looker for coordinators and managers; and Python and SQL automations that removed manual routines from the management team.

CEO · EXPeduca

2017 — 2021

COO (Aug 2017 — Mar 2019)

Ran operations and later the company, part-time, while in university.

SELECTED PROJECTS

AI agents hub · Python, LLMs, SQL, PostgreSQL, Docker

Five areas (HR, treasury, accounting, banking and manufacturing) depended on spreadsheets and manual ERP lookups. I built a platform with 10 agents that read the ERP, the MES and spreadsheets and answer with the data's date and source; AI routes, code executes.

Outcome: On its first run, the accounting agent found 622 wrong records out of 33,000, live, in 12 seconds. The hour-bank agent surfaced 685 negative hours hidden by the consolidated balance.

Task agent on WhatsApp · LLMs, Whisper, WhatsApp, n8n

Plant leaders don't open the task manager on the factory floor. A WhatsApp agent understands text and voice and creates, edits and completes real tasks — echoing what it understood before writing, because transcription hallucinates.

Outcome: Completing a task became photo + caption from the phone, with the evidence the company's rules require. Deadline and owner are parsed in code and checked against the real list; an unknown name becomes a question, never a wrong task.

135 n8n automations · n8n, BigQuery, Graph API, Gotenberg

Reports nobody opened and checks that depended on one person. 2,880 nodes connecting ERP, BigQuery, Microsoft 365 and Google deliver images, PDFs and tasks with deadlines where people already are — WhatsApp and e-mail — every business day, with nobody pressing a button.

Outcome: Every late production order became a task with an owner and a deadline; in the launch week, one plant's backlog of late orders went from 50 to 9. Invoice-receipt reconciliation runs on its own across three plants, without opening a single PDF.

HOW I WORK

Numbers with a date and a source · AI suggests, code decides · Source systems untouched · Cost audited per call.

CONTACT

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SKILLS

AI models and agents

OpenAI · Claude · Gemini · Mistral · LangChain · LangGraph · Whisper

AI-assisted development

Claude Code · Antigravity · VS Code · GitHub · Git

Automation and integration

n8n · APIs REST · Webhooks · WhatsApp · Docker

Data and BI

SQL · Python · Pandas · scikit-learn · Power BI · Microsoft Fabric · Looker

Data engineering and cloud

BigQuery · Google Cloud · PostgreSQL · SQL Server · ETL

EDUCATION AND PUBLICATION

MBA in Data Science and Analytics

University of São Paulo (USP/ESALQ) · 2024 — 2025

Paper accepted at SBPO 2025

57th Brazilian Symposium on Operations Research · 2025

B.Eng. in Mechanical Engineering

Federal University of Ceará · 2016 — 2021